



Introduction to segmentation analysis: Application of biclustering algorithms

Milica Maričić, PhD

University of Belgrade, Faculty of Organizational Sciences

milica.maricic@fon.bg.ac.rs



Agenda

Introduction

Biclustering

Example of market segmentation using biclustering

Algorithms of biclustering and biclustering in R





Introduction

- Clustering - a method of finding similarities in data and grouping data into clusters based on those similarities.
- The amount of available data is increasing (Big data). Do clustering algorithms still "work"?
- Biclustering - represents an upgrade of clustering and implies grouping of data not only by variables, but also grouping by entities that are observed and by variables in relation to which these entities are observed.





Clustering/Biclustering

- Example: We are looking at a data set that contains information about customers and products
- **Clustering** allows grouping by entities (customers who behave in the same way in relation to all products)
- **Biclustering** enables grouping by both entities and variables (customers who behave in the same way in relation to a certain subset of products).





Clustering

- The goal of clustering is to:
 - group data points that are close (or **similar**) to each other
 - identify such groupings (or clusters) in an **unsupervised** manner
- Why unsupervised? There is no information provided to the algorithm on which data points belong to which clusters
- Cluster analysis is used for: Data reduction, Hypothesis generation, Classification of novel entities
- Cluster analysis is criticized for:
 - Being descriptive, atheoretical, and noninferential
 - Always creating clusters, no matter if they actually exist or no
 - Its solution is not generalizable





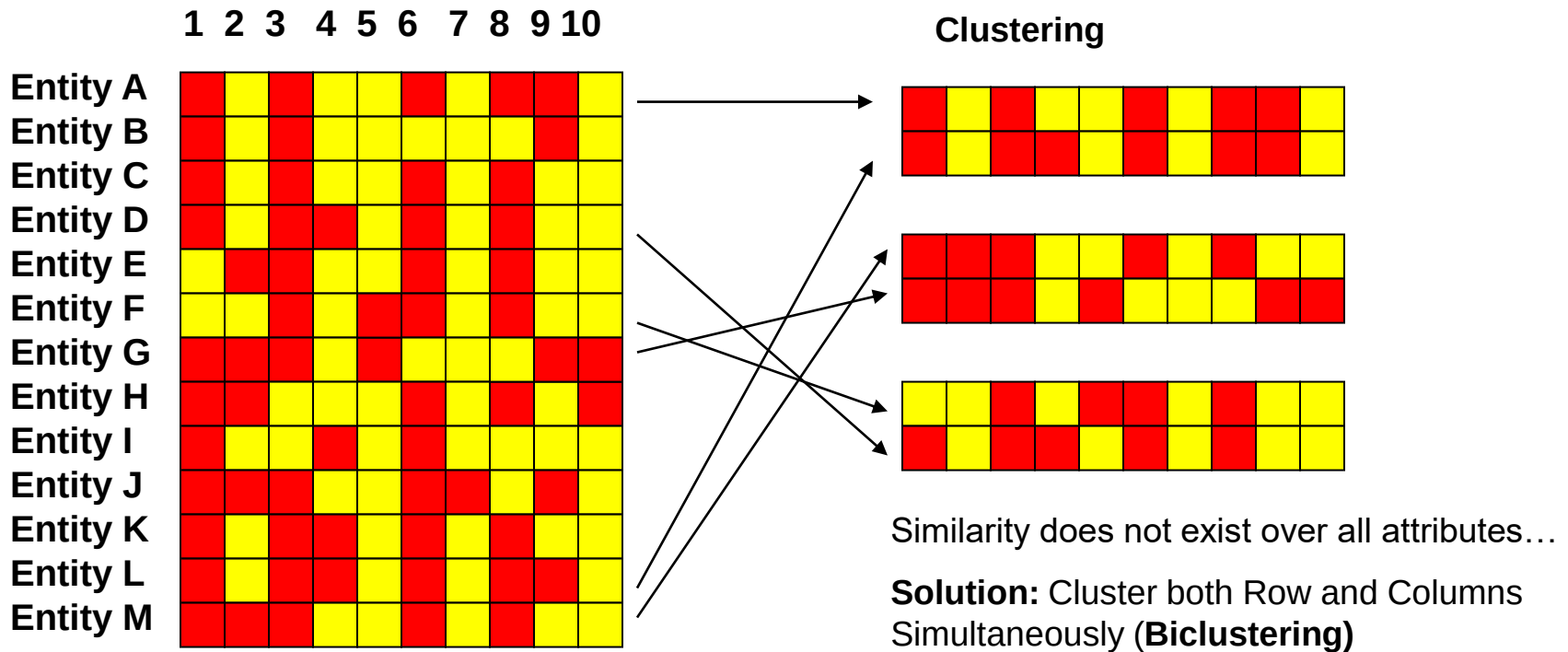
Issues with clustering

- Determining the number of clusters to be formed
- Most algorithms depend on the initial point of execution of the algorithm
- It is sensitive to outliers
- It uses all variables when forming clusters – is there need to use all variables?



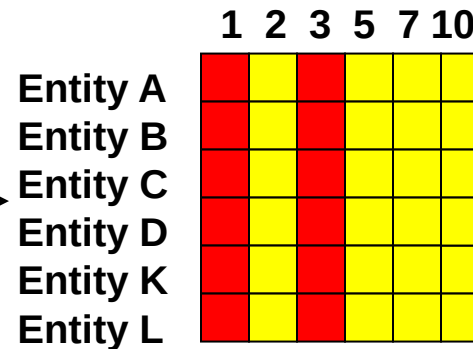
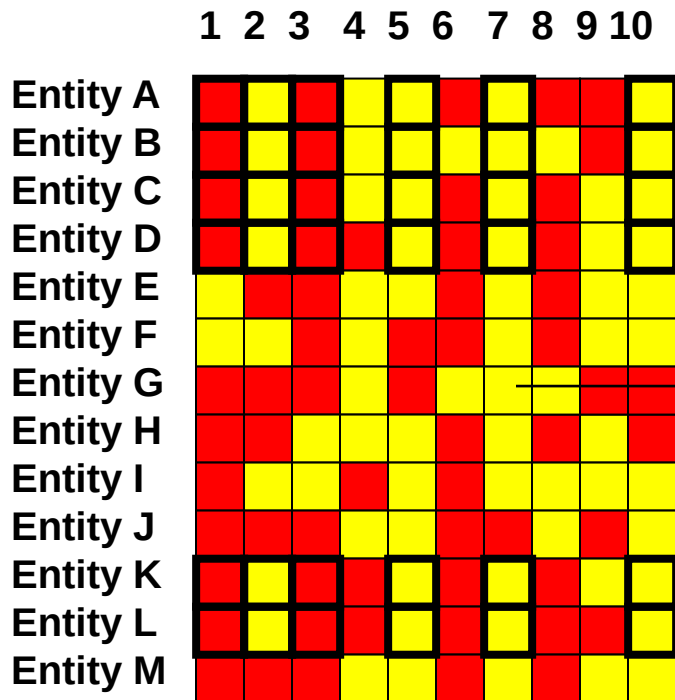


Biclustering vs Clustering





Biclustering vs Clustering



Bicluster {1,2,3,5,7,10}
{A,B,C,D,E,F}





Biclustering - history

- The first biclustering algorithm was proposed by J.A. Hartigan, a professor of statistics at Yale University, back in 1972.
- The term biclustering was introduced in 1996 by Mirkin
- Biclustering's popularity spiked after 2000 when Cheng and Church introduced a biclustering algorithm known as δ -biclustering.
- After that, the interest in biclustering started to grow and the number of methods increased continuously.





Biclustering – areas of application

- Medicine – gene expression
- Marketing – market segmentation
- Tourism – segmentation of tourists based on their needs
- Crisis management – grouping cities and regions
- Quality management – satisfaction/dissatisfaction of consumers
- ...





Biclustering

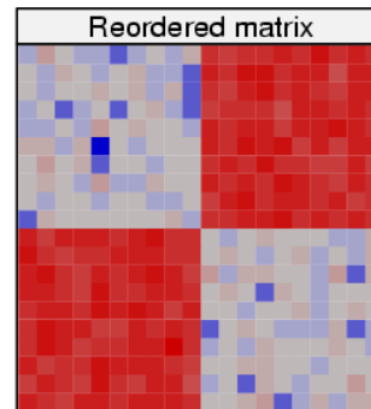
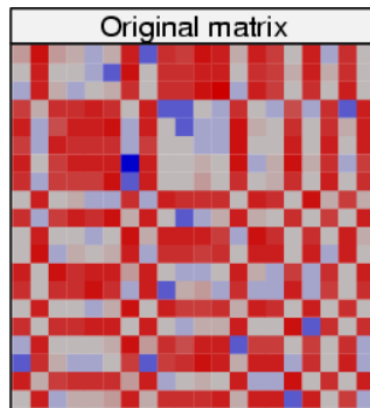
- In biclustering, the focus is on detecting local patterns in the data matrix.
- The goal of the method itself is to identify a group of features that are specific only for a certain subset of entities and form a submatrix of the entire data matrix.
- We call the formed submatrices biclusters.
- A bicluster is a submatrix of the matrix A , with not necessarily adjacent rows and columns, which satisfies certain conditions of homogeneity.





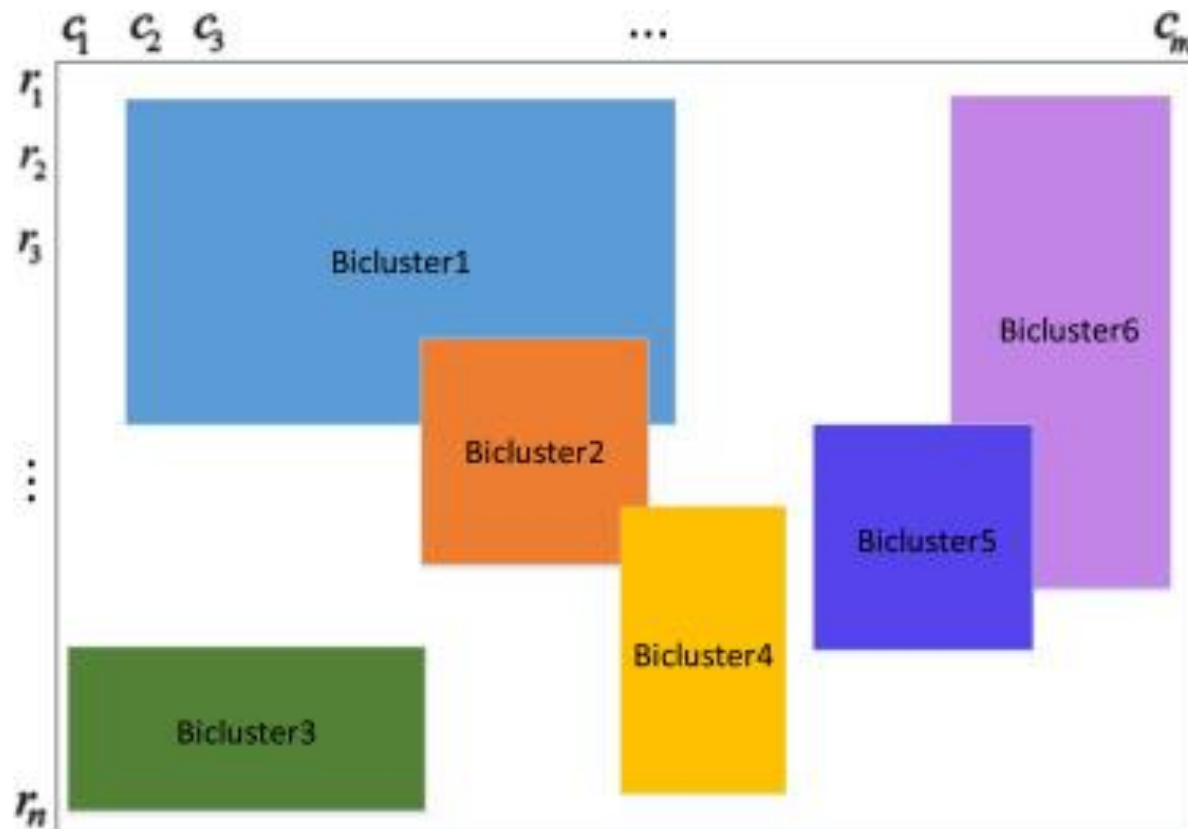
Biclustering

- A biclustering algorithm can be defined as an algorithm whose goal is to find a set of biclusters B such that each bicluster satisfies a certain homogeneity condition.
- The total number of biclusters q depends on the features of the biclustering algorithm that is used, then on the **restrictions** that are applied, the defined **parameters** of the algorithm, as well as on the **structure of the data** set on which the analyses are performed.



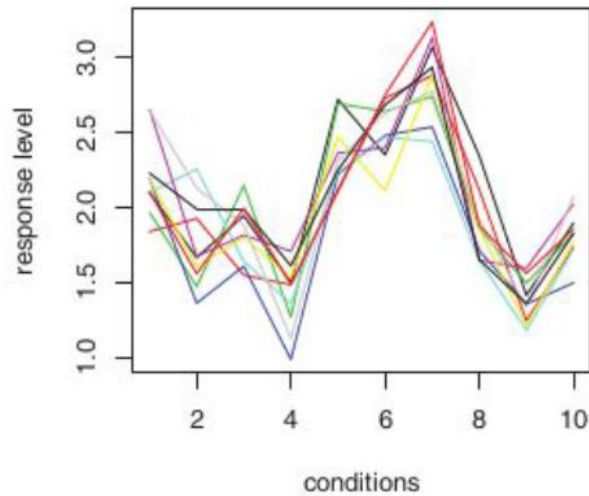


Biclustering

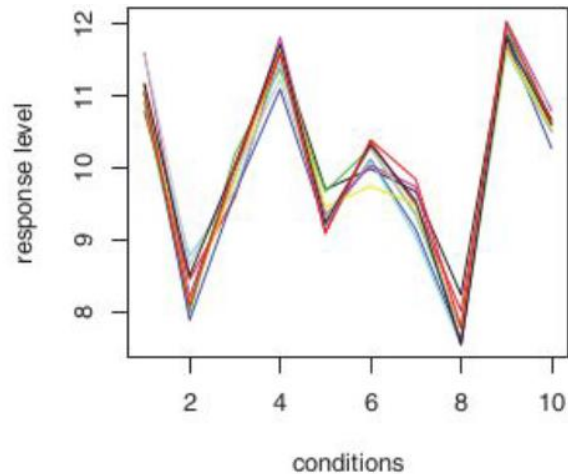




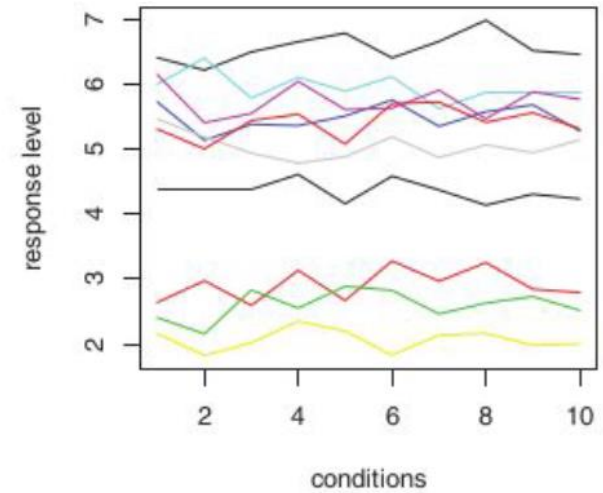
Types of biclusers



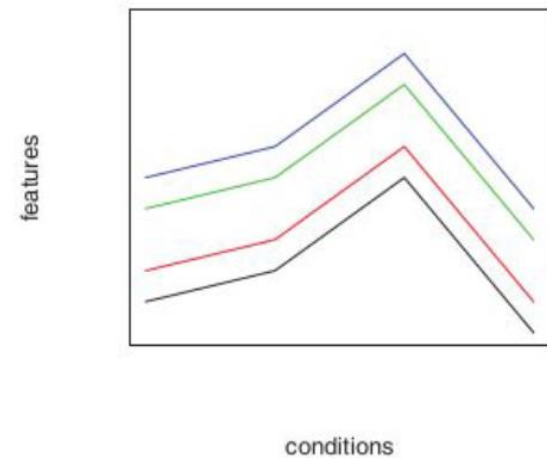
Constant values



Constant values per column



Constant values per row



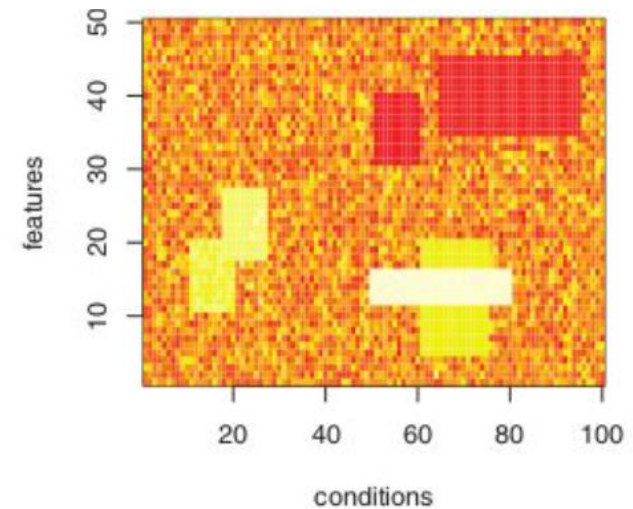
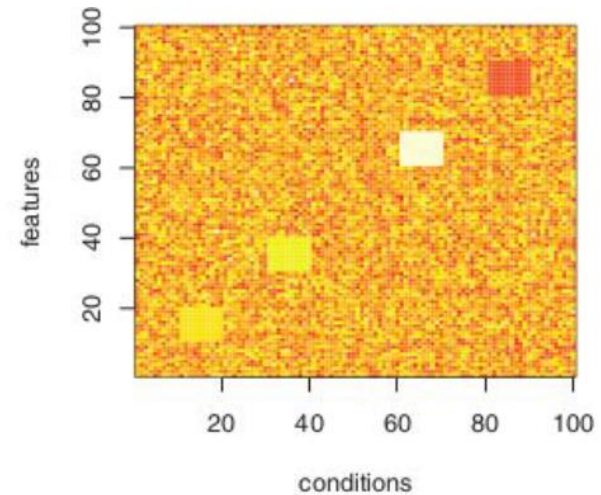
Coherent values





Bicluster configurations

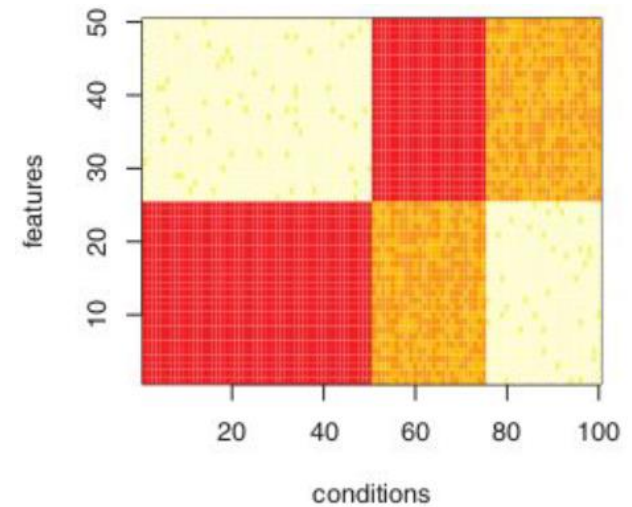
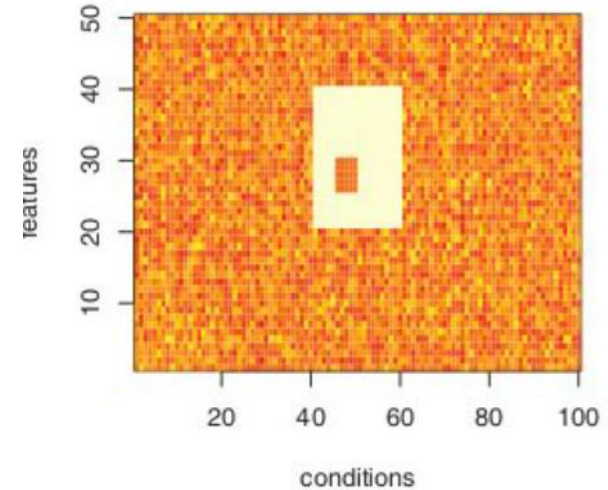
- Exclusive biclusters
- Overlapping biclusters



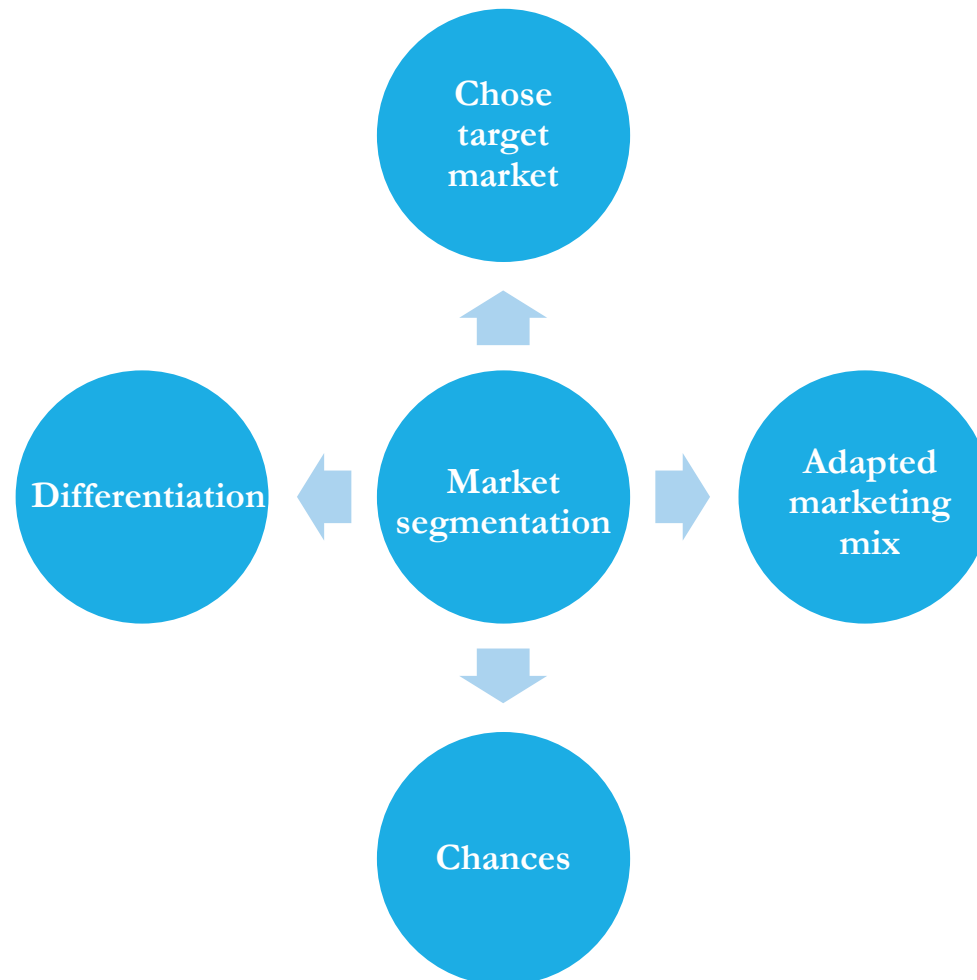


Bicluster configurations

- Hierarchical biclusters
- Checkerboard biclusters



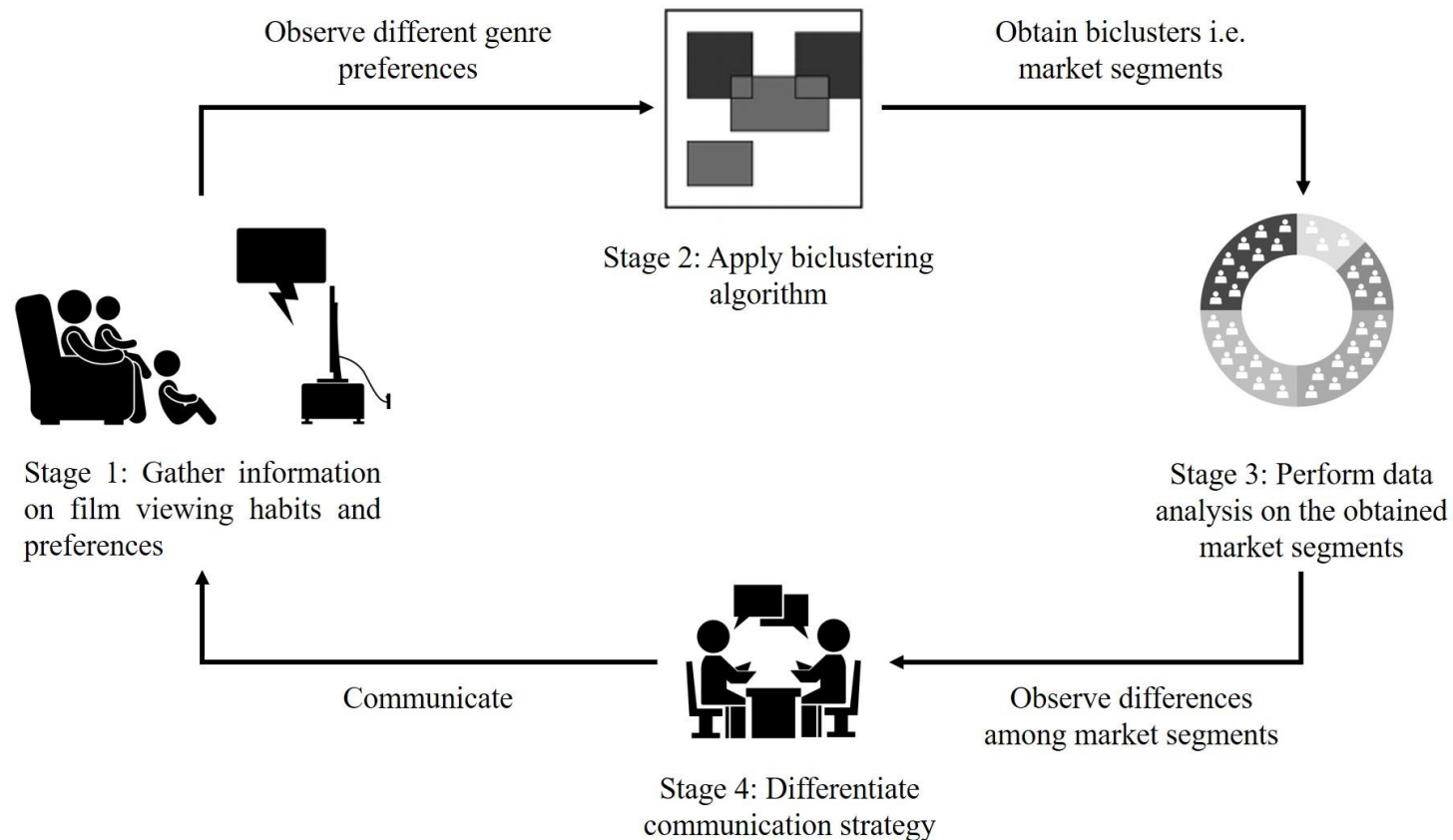
Market segmentation





Example of biclustering in practice - Marketing

- Proposed model:





Example of biclustering in practice - Marketing

- **Sample characteristics:**
- Only millennials, those born between 1980 and 1996, were observed
- A total of 1148 respondents
- Gender: female 832 (72.5%), male 316 (27.5%)
- Average age 28.69, with a standard deviation of 4.39
- 42.5% of respondents have completed university, 37.4% have completed master's degree
- 26.0% of respondents have a monthly income of RSD 50,000-80,000
- 45.6% of respondents are from Belgrade





Example of biclustering in practice - Marketing

- **Biclustering algorithm applied:**
- 11 variables - 11 genres - sci-fi, horror, action, thriller, blockbuster, documentary, comedy, romance, animated, drama and mystery
- Ordinal data, scale 1-5
- The BCQuestord algorithm was used
- Parameterization? There are no rules - trial and error approach
- Metrics that show how good the biclustering results are: the % of the sample that is not classified into any segment, as well as the size of the segments that should be more than 5% of the sample





Example of biclustering in practice - Marketing

- Biclustering results:**

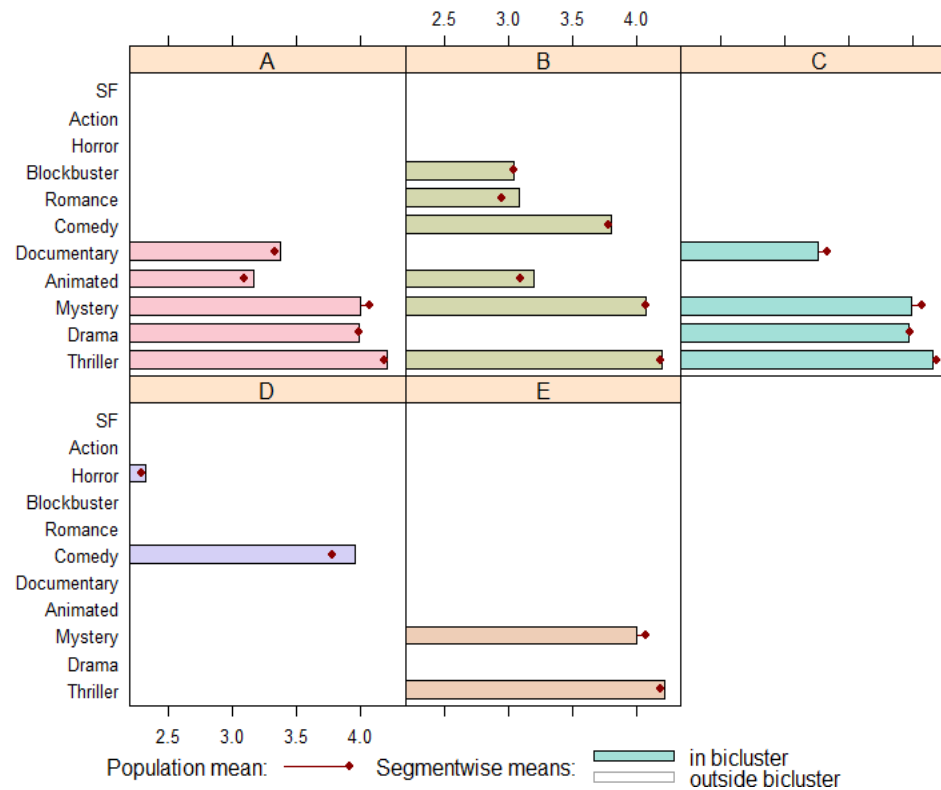
Segment	Size	% sample	% female	Age	Educational attainment
Realists	250	21.77%	77.6%	28.71	Msc (42.8%)
Anti-romantics	104	9.05%	50.0%	28.65	Bsc (61.5%)
Inspectors	286	24.91%	75.2%	28.43	Bsc (43.0%)
Comedians	278	24.21%	77.0%	29.09	Msc (43.5%)
Mystics	113	9.84%	66.4%	28.27	Bsc (44.2%)





Example of biclustering in practice - Marketing

- Biclustering results**





Example of biclustering in practice - Marketing

- **Differences between segments:**

Gender - Chi-square test (35.788, $p < 0.05$) - there is a statistically significant difference in the structures of the segments by gender

Men and women do not prefer the same genres

Age - ANOVA (1.118, $p > 0.05$) - there is no statistically significant difference

There are no differences between younger and older millennials

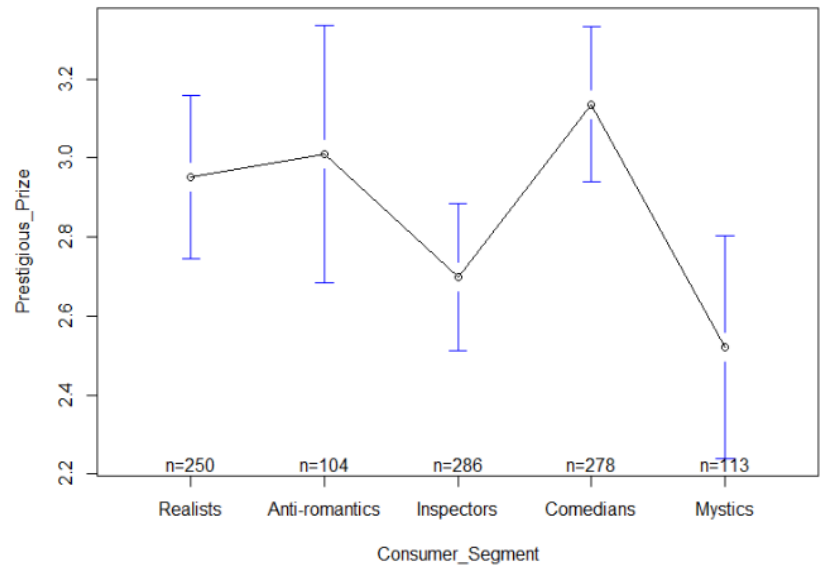
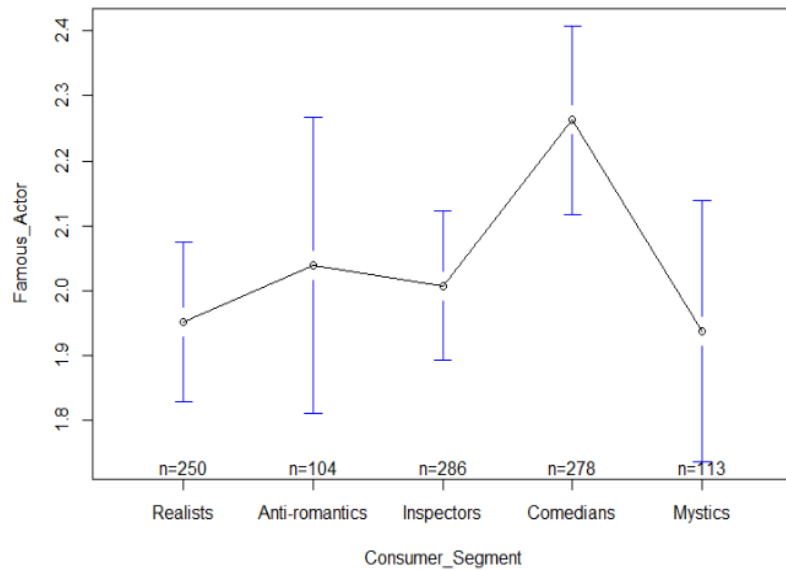
EA - Chi-square test (32.896, $p < 0.05$) - there is a statistically significant difference in the structures of segments according to educational attainment

Depending on EA, respondents prefer different movies



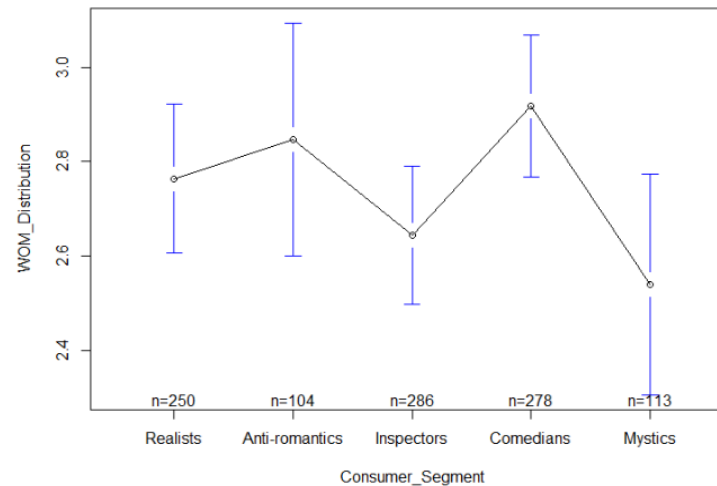
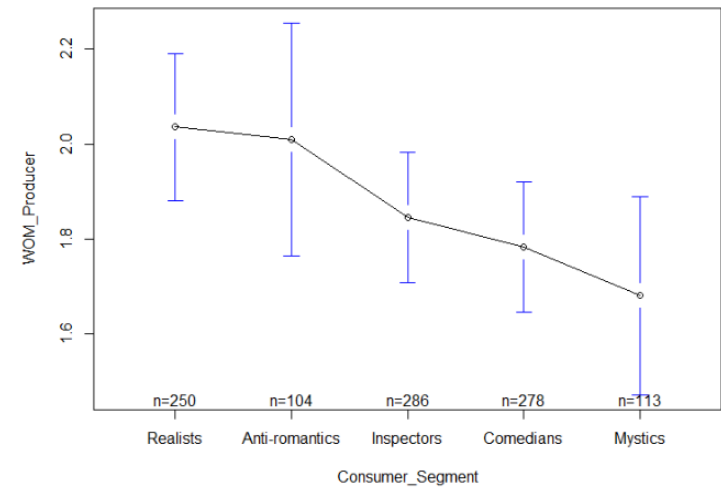
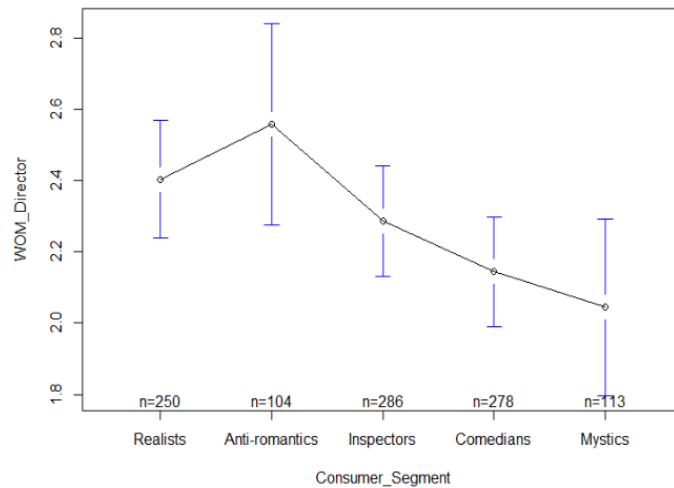


Differences - Product



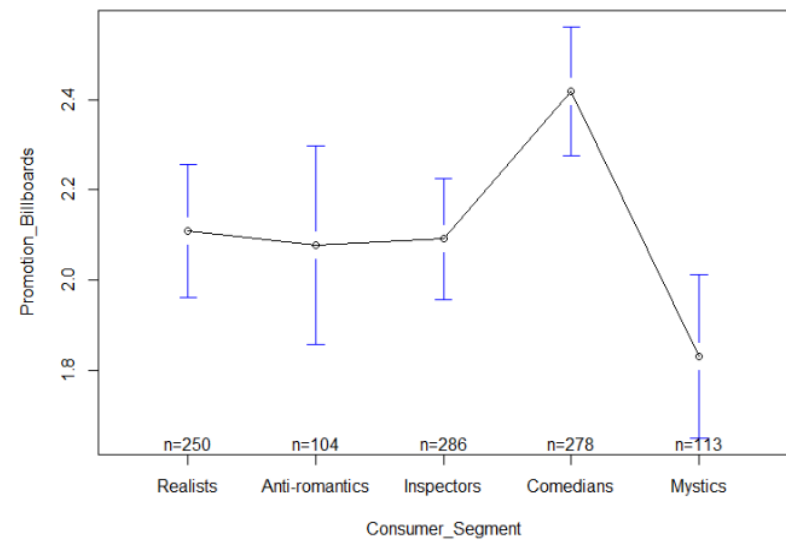
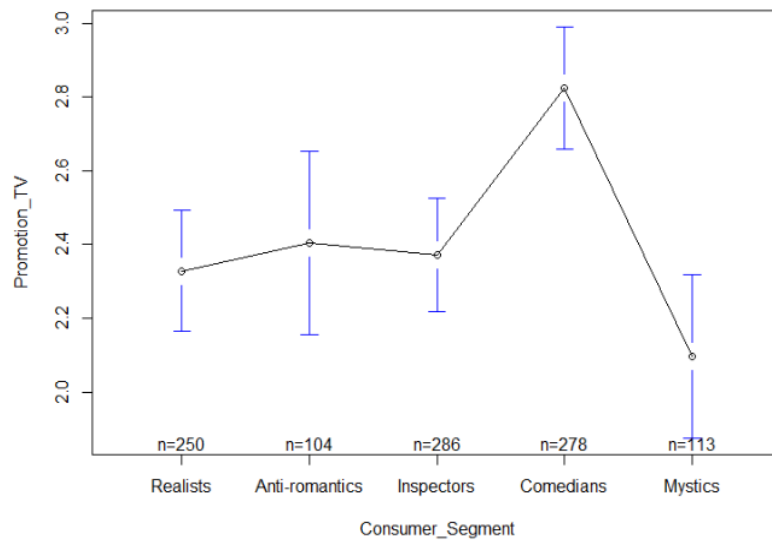


Differences - WOM





Differences - Promotion





Bicluster algorithms

- Iterative row and column clustering combo
 - apply standard clustering methods on the column and row dimensions of the data matrix, and then combine the results to obtain biclusters.
- Divide and conquer
 - algorithm begins with the entire data in one block (bicluster). At each iteration, it finds the row or column that produces the largest reduction in the total “within block” variance by splitting a given block into two pieces.
- Greedy iterative search
 - Greedy iterative search methods are based on the idea of creating biclusters by adding or removing rows/columns from them, using a criterion that maximizes the local gain.
- Exhaustive bicluster enumeration
 - use an exhaustive enumeration of all possible biclusters existent in the matrix.
- Distribution parameter identification
 - assume a given statistical model and try to identify the distribution parameters





Bicluster algorithms

- Different Goals
 - Identify one bicluster.
 - Identify a given number of biclusters.
- Different Approaches
 - Discover one bicluster at a time.
 - Discover one set of biclusters at a time.
 - Discover all biclusters at the same time (Simultaneous bicluster identification)





Bicluster algorithms

- CC (Cheng and Church, ISMB 2000)
- Plaid models (Lazzeroni and Owen, Statistica Sinica 2002)
- SAMBA (Tanay et al, Bioinformatics 2002)
- OPSM (Ben-Dor et al, JCB 2003)
- X-Motifs (Murali and Kasif, PCB 2003)
- ISA - Iterative Signature Algorithm (Ihmels et al, Bioinformatics 2004)
- BiMax (Prelic et al, Bioinformatics 2006)
- BiMine (Ayadi et al, BioData Mining 2009)
- QUBIC (Li et al, NAR 2009)
- FABIA (Hochreiter, Bioinformatics 2010)





Biclustering algorithms covered with the tutorial

- Delta biclustering
- FLOC
- xMotif





Delta biclustering

- Partition-based algorithm
- This algorithm splits the original data matrix into a set of submatrices (biclusters) and uses the variance to evaluate the quality of each bicluster
- Algorithm assumes that there are K biclusters within the data matrix
- In the next step, rows and columns are added or removed to create K biclusters
- Goal: minimize overall variance of the K biclusters





Delta biclustering

- After a bicluster was detected – it is masked with random numbers – so that it cannot be detected again
- Algorithms of adding and removing rows and columns:
 - Single node deletion
 - Multiple node deletion
 - Node addition





Delta biclustering in R

- Below is an example of biclustering in R, with a detailed explanation of the functions and parameters of the functions used
- The example was conducted on a dataset related to a composite indicator of food security

Import the data

```
install.packages(readr)
```

```
library(readr)
```

```
GFSI2021 = read.csv("~Your folder path~/GFSI2021.csv")
```

Get to know your data

Preview the data

```
view(GFSI2021)
```





GFSI index

Country		X1.1	X1.2	X1.3	X1.4	X1.5	X1.6	X2.1	X2.2	X2.3	X2.4	X2.5	X2.6	X2.7	X3.1
1	Algeria	99.0	96.0	63.1	41.1	75	72.4	94.2	46.2	39.1	59.2	49.6	60.3	0	40.3
2	Angola	0.0	23.2	44.2	51.6	50	39.4	26.8	23.1	34.6	85.4	45.9	39.3	50	28.4
3	Argentina	0.0	94.7	60.6	74.4	100	76.4	54.7	31.2	48.1	61.2	69.9	88.2	50	65.7
4	Australia	56.0	99.2	77.1	97.2	100	90.3	81.4	46.7	64.3	12.6	93.0	82.5	50	82.1
5	Austria	88.5	99.2	82.1	72.2	100	92.6	94.2	50.9	76.0	52.4	87.8	84.4	50	79.1
6	Azerbaijan	77.0	100.0	67.8	63.9	100	71.8	48.9	52.1	61.3	74.8	36.2	81.1	50	28.4
7	Bahrain	82.5	99.1	46.6	85.7	100	56.3	75.3	68.1	58.7	98.3	52.1	85.4	0	55.2
8	Bangladesh	71.0	43.8	49.3	56.4	25	52.8	25.6	47.2	58.1	89.5	43.5	77.4	100	1.5
9	Belarus	78.5	100.0	70.4	71.9	100	93.4	55.9	54.5	42.1	52.7	50.8	89.9	50	62.7
10	Belgium	85.5	99.9	83.7	72.2	100	87.7	97.7	41.4	67.7	58.5	87.2	89.5	0	74.6
11	Benin	84.5	18.2	35.8	60.7	25	31.8	43.1	40.8	32.6	72.1	60.9	58.2	50	20.9
12	Bolivia	100.0	91.6	46.3	67.2	50	88.1	48.8	42.2	32.7	27.9	53.4	83.9	100	44.8
13	Botswana	81.5	60.8	34.0	78.4	100	64.7	41.8	83.7	56.6	1.7	76.1	81.1	0	47.8
14	Brazil	54.5	90.2	44.2	74.9	75	86.7	52.4	40.1	36.2	26.9	63.3	53.9	50	71.6
15	Bulgaria	73.0	97.6	64.4	72.2	100	91.7	56.9	47.4	48.3	76.9	69.3	60.3	50	62.7
16	Burkina Faso	80.5	17.6	38.2	60.7	25	34.9	67.4	26.3	27.4	52.7	30.7	76.3	100	23.9
17	Burundi	39.5	3.8	24.1	49.1	25	6.5	7.6	9.6	30.7	51.4	33.4	71.8	50	47.8
18	Cambodia	77.0	63.8	48.5	68.4	100	42.1	22.1	43.8	34.6	88.1	42.7	45.4	100	16.4
19	Cameroon	82.0	49.5	35.1	43.6	25	30.6	65.1	26.6	20.0	92.2	24.8	21.7	0	34.3
20	Canada	88.0	99.5	76.6	62.4	100	85.8	87.2	41.6	72.3	86.1	96.9	81.2	50	77.6

113 countries
26 variables – 1 string, 25 numeric





Application of biclust()

- As there is a string variable in the data set, we have to remove it as we will not use it for biclustering
- Install the package needed to perform delta biclustering

```
# Create a subset
```

```
GFSI2021.use = subset(GFSI2021, select=-c(Country))
```

```
# Install and activate the needed package
```

```
install.packages("biclust")
```

```
library(biclust)
```





Application of biclust()

Create a matrix object

```
GFSI2021.use.matrix <- as.matrix(GFSI2021.use)
```

Guarantee that the same random values are produced each time you run the code
– set random generator seed

```
set.seed(123)
```

Conduct delta biclustering

```
resBCCC <- biclust(GFSI2021.use.matrix, method = BCCC(), delta = 80, alpha =  
1.5, number = 25)
```





Methods covered with biclust()

Biclustering algorithms implemented in the biclust()

- Cheng and Church – BCC()
- Plaid – BCPlaid()
- xMotif – BCXmotifs()
- Spectral – BCSpectral()
- Bimax – BCBimax()
- BCQuest – BCQuest()





Application of biclust()

```
# Show delta biclustering results
```

```
summary(resBCCC)
```

An object of class Biclust

call:

```
biclust(x = GFSI2021.use.matrix, method = BCCC(), delta = 80, alpha = 1.5, number = 25)
```

Number of clusters found: 7

Cluster sizes:

	BC 1	BC 2	BC 3	BC 4	BC 5	BC 6	BC 7
Number of Rows:	39	30	19	10	8	5	2
Number of Columns:	16	11	9	8	6	8	13

Should we redo the analysis and retain a smaller number of biclusters/segments?





Application of biclust()

Basically there is no need to redo the analysis and retain just the 7 biclusters the algorithm detected – the function already did that

```
# Guarantee that the same random values are produced each time you run  
the code – set random generator seed
```

```
set.seed(123)
```

```
# Conduct delta biclustering
```

```
resBCCC <- biclust(GFSI2021.use.matrix, method = BCCC(), delta = 80,  
alpha = 1.5, number = 7)
```

```
# Show delta biclustering results
```

```
summary(resBCCC)
```





Application of biclust()

As we can see, the results are the same – if we called the `set.seed()` function

An object of class Biclust

call:

```
biclust(x = GFSI2021.use.matrix, method = BCCC(), delta = 80, alpha = 1.5, number = 7)
```

Number of clusters found: 7

Cluster sizes:

	BC 1	BC 2	BC 3	BC 4	BC 5	BC 6	BC 7
Number of Rows:	39	30	19	10	8	5	2
Number of Columns:	16	11	9	8	6	8	13

Should we redo the analysis and retain a smaller number of biclusters/segments?





Application of biclust()

```
# Guarantee that the same random values are produced each time you run  
the code – set random generator seed  
set.seed(123)  
  
# Conduct delta biclustering  
resBCCC <- biclust(GFSI2021.use.matrix, method = BCCC(), delta = 80,  
alpha = 1.5, number = 4)  
  
# Show delta biclustering results  
summary(resBCCC)
```

The results and the structure of the biclusters is the same
However, the R object with the results differ





Application of biclust()

List out the values within a particular biculster

```
bicluster(GFSI2021.use, resBCCC, number = 3)
```

```
> bicluster(GFSI2021.use, resBCCC, number = 3)
$Bicluster3
# A tibble: 19 x 9
  x1.2 x1.3 x2.2 x3.3 x3.4 x3.5 x4.1 x4.3 x4.7
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 100 67.8 52.1 90.4 63.9 89.6 68 90.1 73.9
2 90.2 44.2 40.1 89.9 93.7 99.1 67.3 61.4 76.7
3 96.6 52.1 37.1 80.2 75 93.4 79.2 86.4 71.2
4 97.1 54.4 36.2 73 57 92.1 56.9 56.7 72.2
5 89.5 49.8 25 69 61.3 90.8 45.6 60.9 60.3
6 68.9 37.3 33.4 62.2 54.2 89.3 67.5 74.6 51.2
7 78.6 55.1 37.1 54.3 45.1 82 45.5 46.9 69.1
8 92.2 49.4 44.3 93.4 61.9 87.4 76.3 73.8 62.7
9 83.9 47 35.7 55.5 72.4 70.5 52.9 55.5 70.9
10 99.1 75.9 29.6 88.2 94.3 93.6 69.1 63.6 72.7
11 85.9 41.5 47.4 61.6 52.9 82 62.1 70 61.7
12 99.7 85.8 48.3 99.2 100 100 79.1 90.4 68.7
13 95.1 54.2 67.9 80.8 77.4 90.8 68.9 77.2 58.8
14 91.9 51.9 30.1 79.3 63.5 89.1 62.4 66.9 71.1
15 81.7 49.8 43.8 50.9 55.3 89.5 60.7 35.7 61.7
16 99.6 68.3 48.7 93.1 99.4 98.7 65.9 84.1 91.4
17 97.6 64.5 48.5 95.6 75.2 93.1 69.5 85.4 79.3
18 99.8 67.5 36.8 87 81.6 84.3 52.8 67.4 94.2
19 99.5 61.4 21.2 91.1 82 93.3 73.8 68.4 84.2
```





Biclustering plots – Bar chart

Draw a bar chart per bicluster

```
biclustbarchart(GFSI2021.use, resBCCC)
```

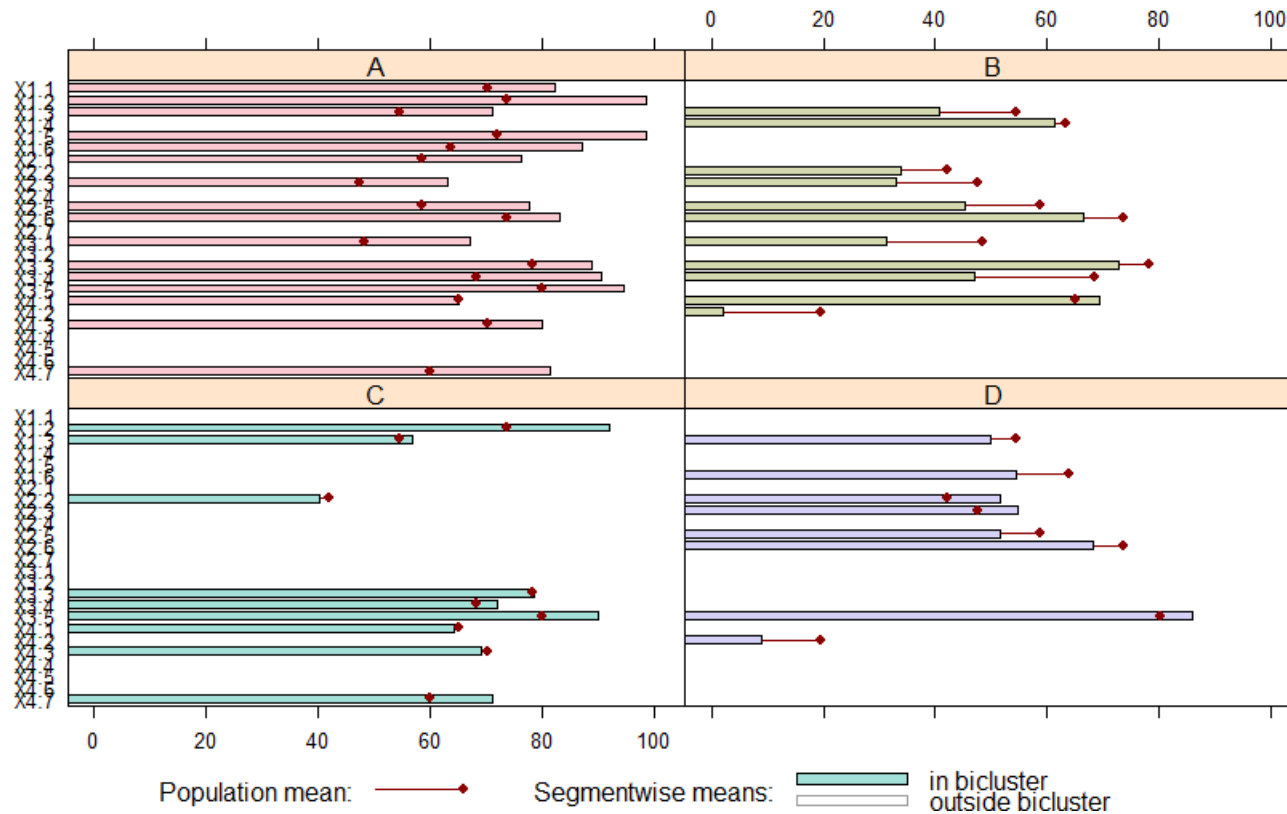
What does this graph show?

- Overall mean per indicator
- Measured mean per indicator per bicluster
- Which indicators are used to form a particular bicluster





Biclustering plots – Bar chart





Biclustering plots – Ordered Bar chart

Object within which the columns are sorted

```
ord <- bicorder(resBCCC, cols = TRUE, rev = FALSE)
```

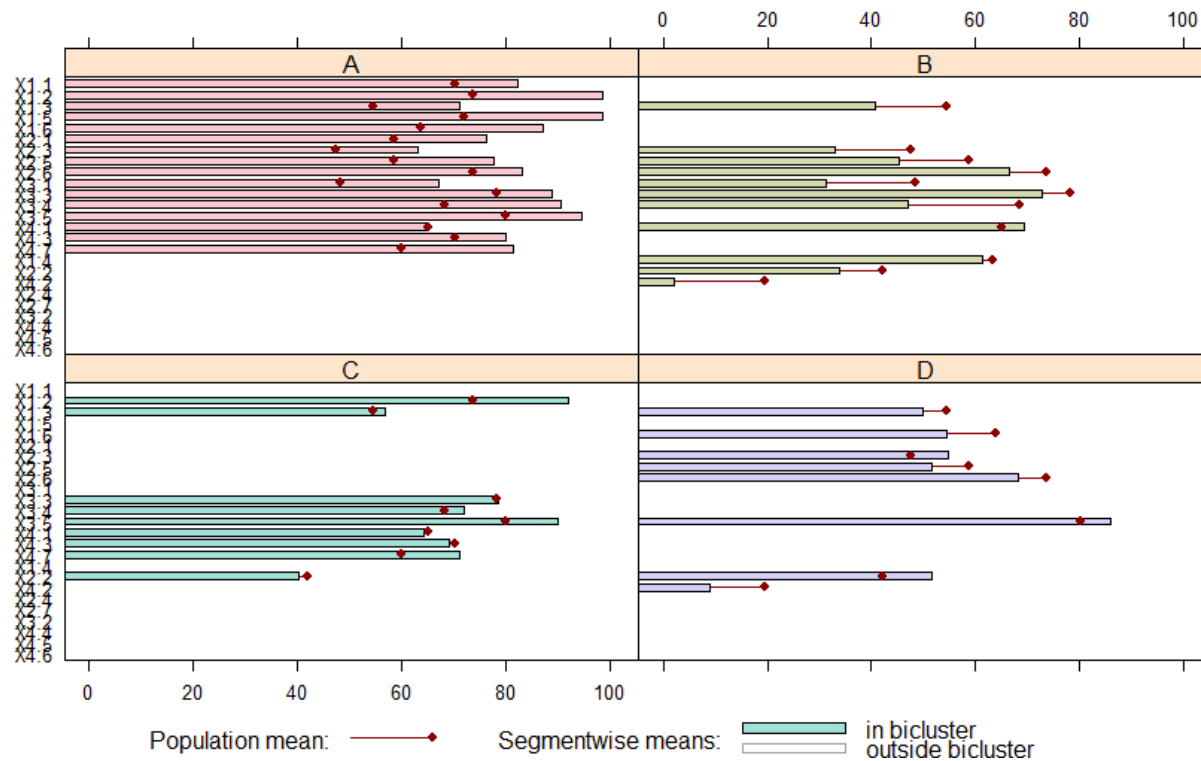
Ordered bar chart per bicluster

```
biclustbarchart(GFSI2021.use, resBCCC, which = ord)
```





Biclustering plots – Ordered Bar chart

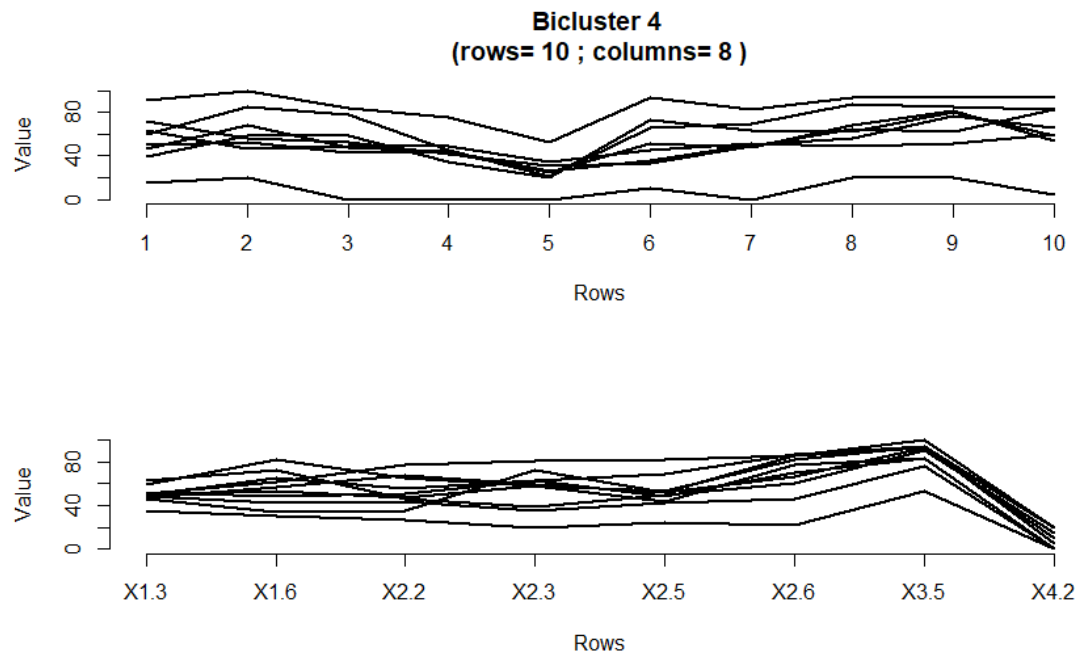




Parallel coordinates plot

Parallel coordinates plot

```
parallelCoordinates(GFSI2021.use.matrix, bicResult = resBCC, number =  
4, compare = FALSE, info = TRUE, plotBoth = TRUE)
```

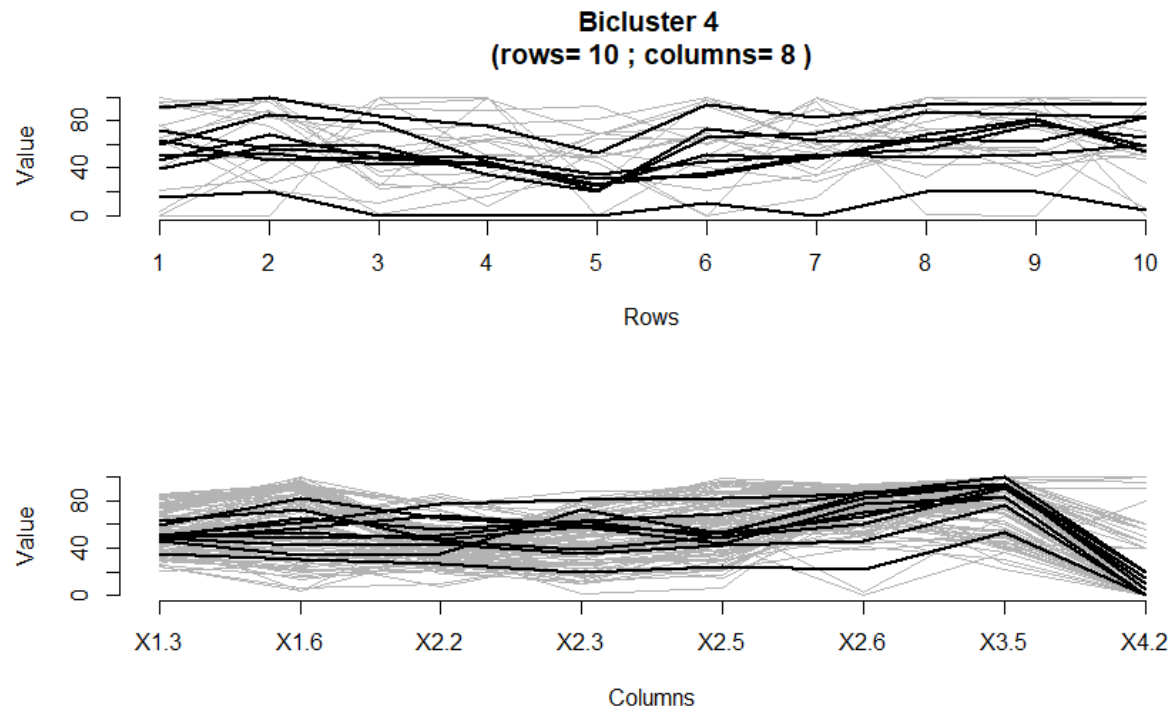




Parallel coordinates plot

Parallel coordinates plot

```
parallelCoordinates(GFSI2021.use.matrix, bicResult = resBCC, number =  
4, compare = TRUE, info = TRUE, plotBoth = TRUE)
```





FLOC algorithm

- FLOC (FLexible Overlapped biClustering)
- Improvement of the delta biclustering algorithm
- Main difference – can be done on datasets with missing data
- Two algorithm phases
 - Phase I: generate K biclusters with a probability p
 - Phase II: Iteratively improve the initial solution measuring information gain. Changes which improve the information gain are preferred





FLOC algorithm in R

```
# Install package BiocManager
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

# Install package BicARE
BiocManager::install("BicARE")

# Call the package BicARE
library(BicARE)
```





FLOC algorithm in R

```
# Set seed for replicability
```

```
set.seed(1234)
```

```
# FLOC algorithm
```

```
resFLOC <- FLOC(GFSI2021.use.matrix, k = 6, pGene = 0.4, pSample = 0.6,  
r = 70, N = 18, M = 9, t = 700)
```

```
# Print the results
```

```
resFLOC
```





FLOC algorithm in R

```
Call :  
FLOC(Data = GFSI2021.use.matrix, k = 6, pGene = 0.4, pSample = 0.6,  
      r = 70, N = 18, M = 9, t = 700)
```

```
Parameters :  
number of biclusters : 6  
residu threshold : 70  
gene initial probability : 0.4  
sample initial probability : 0.6  
number of iterations : 700  
date : Mon Nov 14 12:03:17 2022
```

```
Biclusters :  
      residue volume genes conditions  rowvar  
[1,] 62.53282   486   54           9 249.9395  
[2,] 68.65330   423   47           9 194.2958  
[3,] 67.69475   550   50          11 233.5040  
[4,] 68.15866   468   52           9 183.1583  
[5,] 61.31959   490   49          10 282.6914  
[6,] 69.44558   486   54           9 208.8226
```

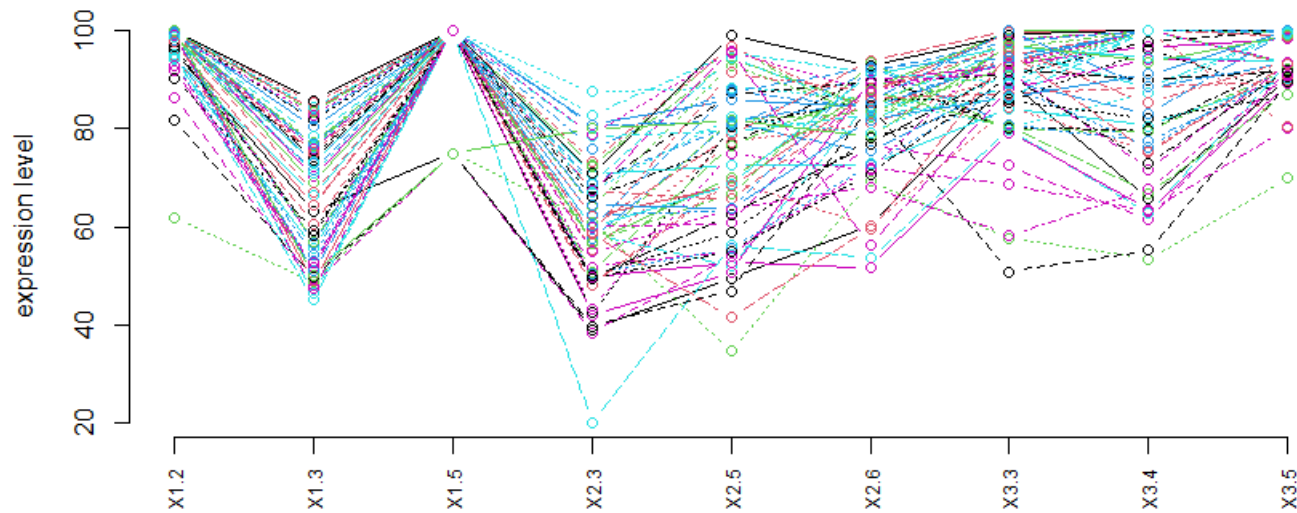




FLOC algorithm in R

```
# Graph of parallel coordinates
```

```
biclusterc(resFLOC, 1, graph = TRUE)
```





Xmotif

- Searches for a submatrix where each row as a similar motif through all columns
- Initially used in gene expression studies – Gene is expressed or no
- Needs binarized data – data must be binary or binarized prior to analysis
- Allows overlapping
- Although it is a decent algorithm it is not used in practice that much
 - Binarization leads to loss of information





Xmotif

- The algorithm Xmotif is more important as its modification is widely used in practice for biclustering nominal and ordinal data – QuestMotif
- QuestMotif has several algorithms
 - BCQuest
 - BCQuestord
 - BCQuestmet





Xmotif biclustering in R

Import the data

```
install.packages(readr)
```

```
library(readr)
```

```
StatFear = read.csv("~Your folder path~/StatFear.csv")
```

Get to know your data

Preview the data

```
view(StatFear)
```





Xmotif biclustering in R

	Gender	Year	SP	MathOne	MathTwo	ProbTheory	q1	q2	q3	q4
1	1	2	1	7	7	5	1	1	1	1
2	1	2	1	7	9	10	1	1	1	1
3	1	2	1	6	6	5	1	1	1	1
4	1	2	1	8	9	10	3	3	2	4
5	1	2	1	9	6	6	1	1	1	4
6	1	2	1	8	8	9	1	1	1	1
7	1	2	1	6	5	5	1	2	2	3
8	1	2	1	10	9	9	1	1	1	1
9	1	2	1	7	9	8	1	1	1	1
10	1	2	1	9	8	8	3	4	1	5
11	1	2	1	8	7	8	2	2	2	2
12	1	2	1	7	6	8	2	3	1	5
13	1	2	1	8	9	8	1	1	1	1
14	1	2	1	10	10	10	1	1	1	1

216 respondents

16 variables – Demographic, Mathematical subjects, Stat Anxiety scale, Stat opinion scale





Xmotif biclustering in R

- Task: Bicluster respondents based on their anxiety and opinion on statistics. All the questions are measured on a Likert scale 1 to 5.
- As all the data is on a ordinal scale – BQuestOrd should be used
- Parametrisation: d – interval around the mean, ns – number of samples, nd – number of subsamples, sd – subsample size





Xmotif biclustering in R

Create a subset

```
StatFear.use = subset(StatFear, select=c(7:16))
```

Create a matrix

```
StatFear.use.matrix <- data.matrix(StatFear.use)
```

Set seed for replicability

```
set.seed(1234)
```

Apply the BCQuestord

```
resBCQ <- biclust(StatFear.matrix, method=BCQuestord(), d=1, ns = 3, nd = 9,  
sd = 4, alpha = 0.01, number = 4)
```

Print the results

```
resBCQ
```





Xmotif biclustering in R

An object of class Biclust

call:

```
biclust(x = StatFear.use.matrix, method = BCQuestord(), d = 1, ns = 3, nd = 9, sd = 4, alpha = 0.01,  
        number = 4)
```

Number of clusters found: 4

First 4 cluster sizes:

	BC 1	BC 2	BC 3	BC 4
Number of Rows:	107	49	33	11
Number of Columns:	4	3	2	6

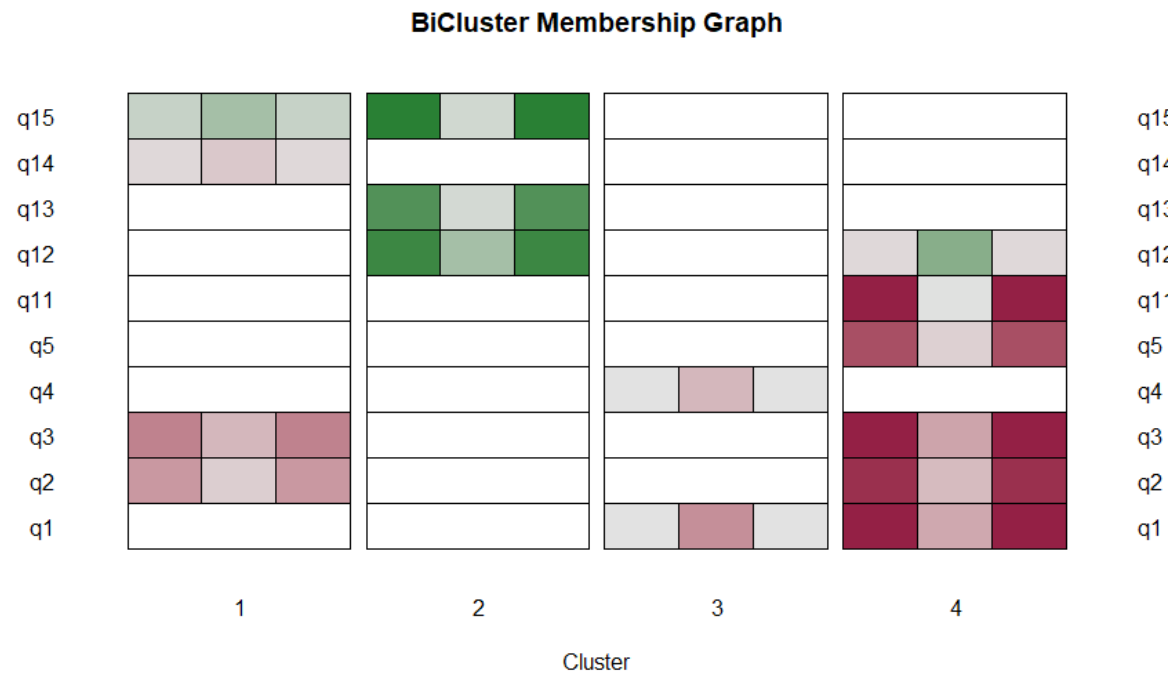




Xmotif biclustering in R

Draws a membership graph for a Bicluster result

```
biclustmember(resBCQ, StatFear.use.matrix)
```



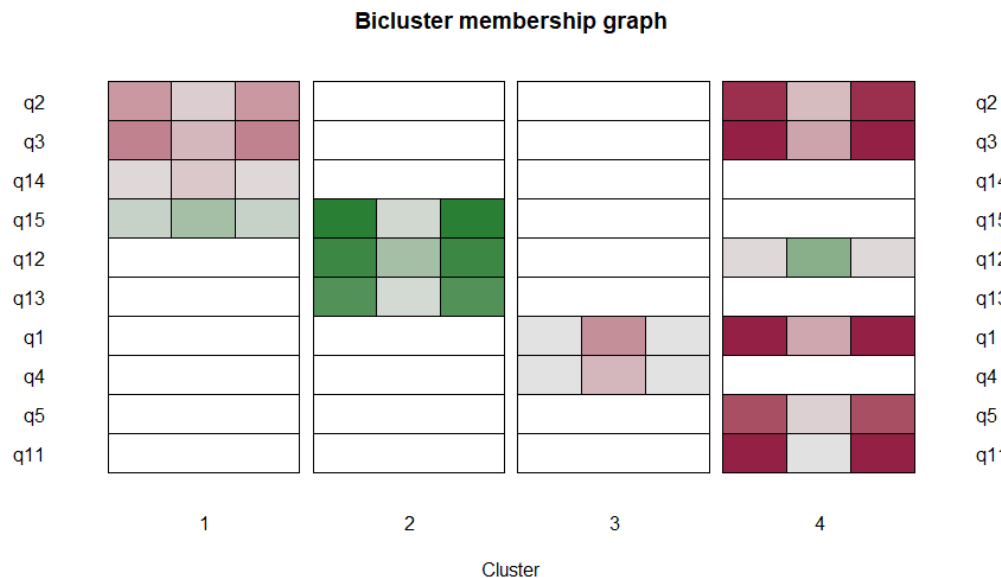


Xmotif biclustering in R

Draws an ordered membership graph

```
ord<-bicorder(resBCQ, cols=TRUE, rev=TRUE)
```

```
biclustmember(resBCQ, StatFear.use.matrix, which=ord, main= "Bicluster membership graph")
```



How can we describe the formed segments?





Why biclustering and not clustering?

- When Clustering algorithms are used
 - Each entity in a given cluster is defined using all the conditions.
 - Each condition in a condition cluster is characterized by the activity of all the entities.
- When Biclustering algorithms are used
 - Each entity in a bicluster is selected using only a subset of the conditions
 - Each condition in a bicluster is selected using only a subset of entities
 - Biclustering identifies groups of genes that show similar activity patterns under a specific subset of the experimental conditions.





Pros and cons of biclustering

Pros

- Indicate which indicators are not necessary for forming a segment
- Vastly used
- Allows the analysis of big data sets and complex phenomena
- There are multiple algorithms available

Cons

- More advanced algorithms – difficult for decision makers to understand
- Parametrization – no single magical answer
- Sensitive to noise and outliers
- No statistical test to prove are the segments statistically significant or no





Introduction to segmentation analysis: Application of biclustering algorithms

Milica Maričić, PhD

University of Belgrade, Faculty of Organizational Sciences

milica.maricic@fon.bg.ac.rs

